

UnifiedGeoRAG: Unified Geolocation and Recommendation via Augmented Generation

413551036 翁祥恩, 313551073 顏琦恩, 314551058 唐文蔚, 314554006 陳乙慈

https://github.com/Tang-Webber/NYCU-IR-2025_Group5_Final

1 Introduction

Geolocation from visual input is a challenging open-world problem: given a single street-view image, the goal is to infer the location in the world where the photo was taken. Unlike standard classification tasks, the set of possible locations is extremely large and constantly expanding, and the visual appearance of different regions varies significantly in architecture, vegetation, signage, and environmental cues. Traditional supervised approaches require large annotated datasets and frequent re-training to incorporate new countries, making them difficult to scale to real-world deployments.

Recent research has explored geolocation using multimodal foundation models. For example, *Img2Loc* (Zhou et al., 2024) demonstrates that image geolocation can be formulated as a retrieval-augmented generation task, where candidate images are retrieved via CLIP embeddings (Radford et al., 2021) and then used as evidence for an MLLM to generate coordinates, which outperforms traditional classification methods that rely on direct image understanding. However, such methods still mainly focus on predicting the location itself and may not provide user-facing, recommendation-style outputs.

In parallel, the multimodal IR community has proposed more effective retrieval pipelines. Works such as *DQU-CIR* (Wen et al., 2024) explore robust CLIP-based visual retrieval for composed image queries, while *IM-RAG* (Yang et al., 2024) introduces multi-round retrieval-augmented generation to allow large language models to reason iteratively using retrieved evidence. These ideas highlight the importance of grounding generation in external retrieved knowledge rather than depending only on learned representations.

Motivated by these developments, our project reframes geolocation as an Information Retrieval (IR) problem and extends it to a landmark recom-



Figure 1: Usage scenario of UnifiedGeoRAG. The system processes a user-provided street-view image to predict the country (e.g., USA) and suggests relevant landmarks, demonstrating its dual capability of geolocation and travel recommendation.

mendation setting. We treat the input image as a visual query and retrieve semantically similar images from a visual knowledge base using CLIP-derived embeddings. A multimodal large language model (MLLM) then integrates the query and the retrieved examples to infer the most likely country. Based on this predicted region, we further retrieve representative landmarks from our curated attraction database and ask the MLLM to generate human-readable descriptions for these landmarks. In this way, the system not only answers “Where is this photo taken?” but also provides “What nearby places might be interesting to visit?”.

Our project therefore addresses two central IR tasks:

- **Visual Retrieval:** retrieving the top-K semantically similar images and landmarks via embedding-based search.
- **Retrieval-Augmented Generation (RAG):** combining retrieved evidence with an MLLM to perform country prediction and generate landmark recommendations with textual descriptions.

2 Related Work

2.1 Image-based Geolocation via Retrieval-Augmented Models

Recent research has shown that geolocation can benefit from retrieval-augmented methods.

Img2Loc (Zhou et al., 2024) demonstrates that image geolocation can be formulated as a retrieval-augmented generation task: visually similar images are retrieved using CLIP embeddings, and a multi-modal large language model (MLLM) uses these retrieved examples to infer geographic coordinates. This work highlights the value of grounding geolocation inference in retrieved visual evidence.

Our project follows a similar retrieval-augmented philosophy, but focuses on country-level prediction instead of coordinate regression, and further extends the task toward user-facing applications such as landmark recommendation.

2.2 Visual Retrieval with CLIP-based Embeddings

Embedding-based retrieval has become a strong baseline for multimodal IR tasks. DQU-CIR (Wen et al., 2024) shows that CLIP embeddings provide robust semantic alignment for tasks such as composed image retrieval, where the system must retrieve an image matching both visual and textual constraints. This line of work supports the idea that CLIP-derived embeddings are effective in large-scale retrieval scenarios.

In our system, we similarly rely on CLIP embeddings and FAISS (Douze et al., 2024) search to retrieve visually similar street-view images and attractions. Unlike composed retrieval, however, our retrieval is used as geographic evidence and as the foundation for downstream recommendation.

2.3 Retrieval-Augmented Reasoning

Retrieval-augmented generation methods aim to improve model reasoning by grounding it with retrieved information. IM-RAG (Yang et al., 2024) introduces a multi-round RAG framework in which a multimodal language model generates internal reasoning steps to decide whether more retrieval is needed before answering. This improves evidence grounding, but requires reinforcement learning and iterative retrieval. In contrast, our system uses a single-round retrieval pipeline, which is simpler and efficient. While ideas from IM-RAG (Yang et al., 2024) are conceptually relevant, our method applies retrieval-augmented reasoning to image-based geolocation and landmark recommendation rather than multi-round decision-making.

3 Dataset

The GeoLocation (Crypticsy, 2024) dataset is a collection of approximately 50,000 street-view screen-

shots from the online game GeoGuessr. Each image is labeled with the country in which the scene is located, making the dataset suitable for evaluating image-based geolocation and classification tasks. The images cover over 150 countries, although the distribution is not uniform; countries such as the United States appear far more frequently, with up to 12,000 images, while many countries have only a few dozen samples. This long-tailed distribution reflects the uneven availability of global street-view data.

The dataset consists of outdoor scenes including roads, buildings, landscapes, and urban environments. Because all images are gameplay screenshots, they include consistent perspective and camera angles, which reduces noise but keeps geographical diversity intact. Labels are provided at the country level, aligning well with our system’s goal of predicting high-level geographic location.

For preprocessing, we resize all images to a fixed resolution for efficient inference and compute CLIP embeddings for each image to build a FAISS index for nearest-neighbor retrieval. Since our method is training-free, no additional annotation or data cleaning is required. The embeddings extracted from this dataset allow us to retrieve visually similar locations and support downstream tasks such as country prediction and landmark recommendation.

4 Baseline

To rigorously evaluate the contribution of our retrieval-augmented and multimodal reasoning components, we established a fundamental baseline using a k-Nearest Neighbors (k-NN) (Steinbach and Tan, 2009) approach based on standard CLIP embeddings. This baseline represents a naive image retrieval method that relies solely on general visual similarity without domain-specific adaptation or complex reasoning capabilities. Specifically, we employed the CLIP ViT-B/32 (Radford et al., 2021) encoder to extract visual features and performed a nearest neighbor search in the embedding space. For the prediction, we simply adopted the geolocation of the single most similar image ($k=1$) from our training set as the predicted location for the query. This method serves as a lower-bound control group, allowing us to quantify the performance gains achieved by introducing the domain-aligned GeoCLIP (Vivanco Cepeda et al., 2023) encoder and the inferential powers of the Multimodal Large Language Model.

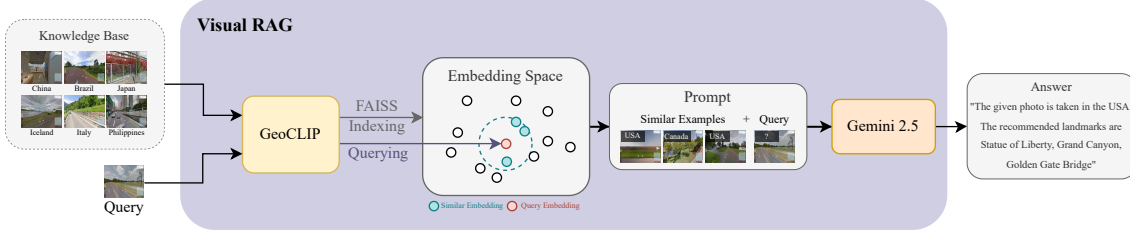


Figure 2: Overview of the proposed UnifiedGeoRAG framework. The pipeline consists of two primary stages: (1) **Visual Retrieval**, where the query image is encoded by GeoCLIP to retrieve top- K visually similar examples from the FAISS index; and (2) **Multimodal Generation**, where these retrieved examples—serving as visual anchors with ground-truth metadata—are combined with the query into a composite prompt. This enables the MLLM (Gemini 2.5) to ground its reasoning in the provided evidence, resulting in accurate country prediction and relevant landmark recommendations.

5 Main Approach

5.1 System Architecture

To address the challenge of open-set geolocation, we developed a framework named **UnifiedGeoRAG**. This system integrates dense visual retrieval with the reasoning capabilities of multimodal large language models. The architecture comprises two primary stages: a Visual Retrieval Module that identifies geographically relevant context, and a Multimodal Generation Module that synthesizes this information to predict the final location. This structured approach allows the system to ground its predictions in concrete visual evidence from a reference knowledge base rather than relying solely on internal model knowledge.

5.2 Visual Retrieval Module

The retrieval component is responsible for identifying relevant reference images for a given query. We utilized two distinct visual encoders to extract feature embeddings: the standard CLIP ViT-B/32 model (Radford et al., 2021) and GeoCLIP (Vivanco Cepeda et al., 2023). GeoCLIP (Vivanco Cepeda et al., 2023) is specifically fine-tuned on street-view imagery to capture location-specific features such as vegetation patterns and architectural styles.

To enable efficient similarity search over the dataset, we constructed a vector index using the FAISS (Douze et al., 2024) library. We specifically employed the IndexFlatL2 structure to ensure exact nearest neighbor search based on Euclidean distance. During the inference phase, the system encodes the input image and retrieves the top- K most similar images from the index. These retrieved images, accompanied by their ground-truth meta-

data (i.e., known coordinates and country labels), serve as the factual basis and visual context for the downstream generation task.

5.3 Multimodal Generation Module

The generation component leverages the retrieved context to perform precise geolocation through reasoning. We adopted the Gemini 2.5 Flash Image model (Comanici et al., 2025) as our core Multimodal Large Language Model (MLLM). The system employs a dynamic few-shot prompting strategy where the input is constructed specifically for each query.

The prompt structure consists of three distinct parts:

- **Task Instruction:** A direct directive asking the model to predict the geographic location based on the provided evidence.
- **Contextual Examples:** The set of K retrieved images is presented to the model as similar examples, each accompanied by its ground-truth coordinates and country name.
- **Visual Query:** The target image is appended at the end of the sequence.

By processing this composite multimodal input, the model analyzes the visual similarities between the query and the retrieved examples. It then infers the most probable location by grounding its reasoning in the provided reference data. To ensure the predictions are computationally actionable, we define a strict expected output format within the system instructions. The model is directed to generate a structured response consisting of three specific predictive fields: the estimated Latitude, Longitude, and the Country name. This standardization allows

Table 1: Quantitative comparison of the proposed method with several baselines. The table reports the country prediction accuracy and mean distance error (MDE) for each approach. Our Visual RAG method demonstrates superior classification performance compared to traditional supervised learning (DenseNet121, Huang et al., 2017) and zero-shot MLLM baselines. Note: For the overall comparison, we selected the optimal K value for each method based on validation performance (K=10 for KNN and K=3 for RAG).

Method	Country Accuracy (%)	Mean Distance Error (km)
DenseNet121	34.23	-
KNN w/ CLIP (Baseline)	34	2979.32
KNN w/ GeoCLIP	56	1311.11
Gemini	60	841.67
UnifiedGeoRAG (Ours)	64	1455.52

the system to programmatically parse the reasoning results, enabling the calculation of geodetic distance errors against the ground truth.

By integrating the GeoCLIP vector embedding and offering a targeted set of the top-K relevant context images (K=3), the Multimodal Large Language Model (MLLM) is effectively grounded, leading to more accurate country predictions for the query image. We have named our georetrieval method UnifiedGeoRAG.

6 Evaluation Metric

To evaluate the performance of our geolocation and recommendation system, we use a combination of quantitative and qualitative metrics. These metrics measure not only classification accuracy but also retrieval quality and the effectiveness of the recommended landmarks.

6.1 Country Prediction Accuracy

We evaluate the classification performance using standard accuracy, defined as the proportion of test images whose predicted country matches the ground-truth country.

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N 1(\hat{y}_i = y_i) \quad (1)$$

where N is the number of test images, $\hat{y}_i$ is the predicted country, and y_i is the ground-truth label.

6.2 Rank@K

To measure retrieval performance, we use Rank@K, which evaluates whether the correct country appears within the top-K retrieved nearest neighbors from the FAISS index.

$$\text{Rank@K} = \frac{1}{N} \sum_{i=1}^N 1(y_i \in \text{TopK}(x_i)) \quad (2)$$

This metric captures how effectively the CLIP embedding space retrieves meaningful visual neighbors that can support downstream reasoning.

6.3 Mean Distance Error

Since our system performs country-level prediction, we evaluate geographic accuracy by calculating the distance between the representative center points of the predicted and ground-truth countries. Specifically, for each test image, we identify the geographic centroid (latitude and longitude) of the predicted country and the ground-truth country. We then compute the distance between these two center points. The Mean Distance Error (MDE) is defined as the average distance across the test set:

$$\text{MDE} = \frac{1}{N} \sum_{i=1}^N d(\text{center}(\hat{y}_i), \text{center}(y_i)) \quad (3)$$

where $\text{center}(\cdot)$ returns the geographic centroid of a given country, and $d(\cdot)$ computes the haversine distance (in kilometers) between the two coordinates. This metric provides a continuous measure of error that accounts for the geographic proximity of the predicted region to the true location, even when the exact country label is incorrect. For context, the typical geographic distance between neighboring countries is 200–700 km, the scale of a cultural region is roughly 1,500 km, and the distance across a continent is approximately 2,300 km.

7 Results & Analysis

Our experimental results demonstrate that the proposed Visual RAG framework significantly outperforms traditional classification approaches, as shown in Table 1. We compared our method against a standard DenseNet121 (Huang et al., 2017) classification model trained on the same Geoguessr

Table 2: Geographic precision analysis at different distance thresholds. This table evaluates the localization accuracy across varying scales, from city-level (25 km) to continent-level (2500 km). While the zero-shot Gemini model shows strong performance in broad regional classification, UnifiedGeoRAG significantly improves fine-grained localization, achieving higher accuracy at the city level (25 km) by effectively leveraging retrieved visual context.

Method	Distance (Accuracy [%] @ km)			
	City (25 km)	Region (200km)	Country (750 km)	Continent (2500 km)
Gemini	6	42	76	94
RAG (GeoCLIP, K=3)	28	48	72	84
RAG (GeoCLIP, K=10)	42	46	66	90

(Crypticsy, 2024) dataset, which achieves a country accuracy of approximately 34.23%. In contrast, our best-performing configuration, using GeoCLIP (Vivanco Cepeda et al., 2023) for retrieval and Gemini 2.5 for reasoning with $K = 3$, achieved a country-level prediction accuracy of 64.0%. This substantial margin highlights the superiority of leveraging foundation models for open-world geolocation tasks over standard supervised learning pipelines.

We have observed a minor performance drop in our method’s effectiveness within MDE. This is primarily due to our reliance on a dataset classified by country rather than using more precise geolocation coordinates. As a result, the MDE is calculated based on country centroids instead of exact locations, which inevitably constrains its geographical accuracy. Nevertheless, our method continues to perform within an acceptable range, remaining consistent with the same cultural region.

Furthermore, we analyzed the relationship between the country accuracy and MDE, observing an interesting trade-off as shown in Table 2. While the Zero-shot Gemini (Comanici et al., 2025) setting yielded the lowest Mean Distance Error of 841 km, the Retrieval-Augmented Generation approach provided higher classification accuracy in specific configurations. The RAG system with GeoCLIP ($K = 3$) achieved an accuracy of 64%, surpassing the Zero-shot Gemini (Comanici et al., 2025) performance of 60%. This suggests that while the internal knowledge of the MLLM is highly effective for approximate localization, the retrieval of visually similar examples provides crucial evidence that helps the model correctly identify the specific country, effectively grounding the prediction in concrete visual data.

8 Error Analysis

We conducted an ablation study to investigate the impact of different components and hyperparameters on system performance.

Table 3: Ablation study on visual encoders using the k-NN baseline ($K = 10$). This table compares the Mean Distance Error (MDE) of Standard CLIP versus GeoCLIP, highlighting the effectiveness of domain-specific pre-training.

Method	Encoder	Mean Distance Error (km)
KNN	Standard CLIP	2979.32
	GeoCLIP	1311.11

Table 4: Impact of visual encoders on RAG performance ($K = 3$). This table compares the country-level classification accuracy of the RAG system when equipped with Standard CLIP versus GeoCLIP.

Method	Encoder	Country Accuracy (%)
RAG	Standard CLIP	58
	GeoCLIP	64

First, we analyzed the effect of the visual encoder, as illustrated in Table 3. The results consistently showed that GeoCLIP (Vivanco Cepeda et al., 2023) outperforms the standard CLIP encoder across both k-NN and RAG settings. For instance, in the k-NN voting ($K = 10$) experiment, GeoCLIP (Vivanco Cepeda et al., 2023) reduced the Mean Distance Error to 1311 km compared to 2979 km for CLIP (), confirming that domain-specific pre-training captures more relevant geographic features such as vegetation and architectural styles.

Second, we evaluated the contribution of the reasoning module by comparing simple k-NN retrieval against the full RAG pipeline, as shown in Table 4. The RAG approach generally yielded higher classification accuracy than k-NN using the same encoder. For example, with $K = 3$ and GeoCLIP (Vivanco Cepeda et al., 2023), RAG achieved 64% accuracy compared to 42% for k-NN voting. This indicates that the MLLM successfully synthesizes the retrieved visual context to make more informed decisions rather than simply relying on feature similarity.

Table 5: Impact of retrieval set size (K) on RAG performance. The results indicate that $K = 3$ offers the optimal balance between providing sufficient context and minimizing visual distractions, achieving the highest country prediction accuracy.

Method	K Value	Country Accuracy (%)	Mean Distance Error (km)
RAG (GeoCLIP)	1	50	1732.03
	3	64	1455.52
	5	56	1416.33
	10	54	1321.75

Finally, we examined the influence of the number of retrieved context images, denoted as K , as shown in Table 5. Our experiments revealed that $K = 3$ represents an optimal balance for this task. Increasing K to 10 resulted in a decrease in accuracy to 54%. We hypothesize that a larger K introduces irrelevant images or visual distractors into the context window, which may confuse the model and lead to hallucinations. Conversely, a very small K might not provide sufficient evidence. Therefore, providing a focused set of three highly relevant examples maximizes the reasoning capability of the model.

9 Future Work

Currently, our retrieval system relies on an embedding space constructed solely from visual features extracted from the images. A promising direction for future research is to enrich this embedding space by incorporating additional multimodal information associated with the reference images. We plan to explore methods to integrate geographic metadata, such as latitude and longitude coordinates and country labels, as well as textual descriptions of the scene (e.g., climate conditions, vegetation, or architectural details) into the embedding learning process. By fusing these diverse data sources into a unified embedding space, we hypothesize that the system can capture more discriminative features, thereby improving the accuracy of country-level geolocation predictions.

Contribution of each member

Student ID	Name	Contribution
413551036	翁祥恩	Methodology Design, Paper Writing
313551073	顏琦恩	Methodology Design, Paper Writing
314551058	唐文蔚	System Implementation
314554006	陳乙慈	Methodology Design, Paper Writing

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